

RESEARCH ARTICLE

Strong linkage between plant and soil fungal communities along a successional coastal dune system

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One sentence summary: Strong correlation between vegetation and soil fungal communities, similar responses to soil properties, and increasing diversity of plant nutrient-acquisition strategies were revealed along a coastal dune successional and edaphic gradient.

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ABSTRACT

Complex interactions between plants and soil microorganisms drive key ecosystem and community properties such as productivity and diversity. In nutrient-poor systems such as sand dunes, plant traits and fungal symbioses related to nutrient acquisition can strongly influence vegetation dynamics. We investigated plant and fungal communities in a relic foredune plain located on an archipelago in Québec, Canada. We detected distinct communities across the edaphic and successional gradient. Our results showed a clear increase in plant species richness, as well as in the diversity of nutrient-acquisition strategies. We also found a strong correlation between aboveground vegetation and soil fungal communities, and both responded similarly to soil physicochemical properties. Soil pH influenced the composition of plant and fungal communities, and could act as an important environmental filter along this relic foredune plain. The increasing functional diversity in plant nutrient-acquisition strategies across the gradient might favor resource partitioning and facilitation among co-occurring plant species. The coordinated changes in soil microbial and plant communities highlight the importance of aboveground–belowground linkages and positive biotic interactions during ecological succession in nutrient-poor environments.

Keywords: 454-sequencing; coastal sand dunes; plant–soil feedback; soil fungal community; plant nutrient acquisition; ecological succession

INTRODUCTION

There is growing recognition that interactions between plants and soil microorganisms drive important community and ecosystem properties such as diversity and productivity (Bever et al. 2010; van der Putten et al. 2013; Barberán et al. 2015). In

particular, soil fungi—as plant symbionts or as major regulators of decomposition, carbon sequestration and soil respiration—are key drivers of vegetation dynamics (Wardle et al. 2004; Boddy, Frankland and van West 2007; van der Heijden, Bardgett and van Straalen 2008; Bever et al. 2010). Along with other

microorganisms, they decompose dead organic material, ensuring the recycling of essential nutrients in soils and thus governing nutrient availability (Boddy, Frankland and van West 2007; van der Heijden, Bardgett and van Straalen 2008). Soil properties, particularly those related to nutrient availability, strongly influence plant growth and community composition (Lambers et al. 2008; van der Heijden, Bardgett and van Straalen 2008).

Nutrient acquisition by plants usually involves specialized root structures or symbiotic interactions with fungi (Lambers et al. 2008; Zemunik et al. 2015). These nutrient-acquisition strategies exist in the majority of plants species—over 90% of plant species have specialized features for nutrient uptake (Lambers et al. 2008). Some plants developed specialized strategies such as parasitism or carnivory, or display a specialized root morphology such as the brush-like cluster roots, while others form symbiotic associations with nitrogen-fixing bacteria such as *Rhizobia* or *Frankia* or with mycorrhizal fungi. Arbuscular mycorrhizas (AM) are by far the most common strategy, occurring in more than 80% of vascular plant species (Brundrett 2009). This symbiosis is the result of the colonization by Glomeromycota fungal hyphae of the cortex of plant roots, where they form typical structures such as arbuscules, vesicles and, sometimes, intraradical spores. Fungal hyphae extend beyond the roots and take up nutrients, in particular phosphorus (P), which is translocated through the hyphae into arbuscules where it is exchanged for carbon (Smith and Read 2008). There exist other types of mycorrhizal symbioses: ectomycorrhizas involve N uptake and include a wide range of fungi in the Basidiomycota and Ascomycota in association with trees and shrubs; ericoid mycorrhizas are formed by specific soil ascomycetes partnered with Ericaceae; and orchid mycorrhizas are key to orchid seed germination (Smith and Read 2008). Despite their ecological importance, soil fungal communities, and microbes in general, are often not considered in vegetation studies. However, a better understanding of plant–soil feedbacks and soil fungal community dynamics is imperative to improve our understanding of ecological processes such as succession (Wardle et al. 2004; Kardol and Wardle 2010). Aboveground and belowground components of ecosystems have traditionally been studied separately, but they must be studied together to understand ecosystem functioning (Wardle et al. 2004).

In nutrient-poor systems such as sand dunes, plant traits and fungal symbionts related to nutrient acquisition are expected to strongly influence vegetation dynamics and assembly (Lambers et al. 2008). As such, early work indicated close association between the development of vegetation and the abundance and identity of cultivable soil microbes (bacteria and fungi) along a sand dune successional gradient (Webley, Eastwood and Gimingham 1952). Yet, little is known about these belowground traits and the specific mechanisms linking above- and belowground biota remain at least partially unresolved (Barberán et al. 2015). Moreover, these feedbacks were only rarely considered in the context of ecological succession (Kardol et al. 2013; Walker and Wardle 2014). Coastal sand dunes are dynamic ecosystems presenting a sequence of vegetation spanning from low plant diversity in upper beach to fixed dunes with diverse vegetation. These habitats are typically characterized by different successional stages with distinct plant communities (Cowles, 1899; Olson 1958; Lichter 1998; Maun 2009).

Here, we investigated the links between fungal and plant communities across a successional and edaphic gradient in a relic foredune plain (Giles and King 2001). Building on our previous studies along this sequence where we investigated changes in soil microbial community composition (Roy-Bolduc

et al. 2015), we explored linkages between soil microbial and plant communities across the dune system. Specifically, our study aimed to answer the following questions. Does the relative abundance of different plant nutrient-acquisition strategies vary along this edaphic and successional gradient? How does above-ground vegetation relate to belowground fungal communities in terms of diversity and community structure? Are changes in plant and/or fungal community structure related to variations in soil characteristics (e.g. pH, nutrients)? To address these questions, we performed a comprehensive sampling of soils and aboveground vegetation along a coastal dune system forming a successional and edaphic gradient (Giles and King 2001; Roy-Bolduc et al. 2015). Vegetation cover by species was recorded at each sampling site and belowground fungal communities were characterized using 454-sequencing. We studied patterns of plant and fungal diversity, and of plant nutrient-acquisition strategies. We also related plant and fungal community structure using co-inertia analysis and explored the effect of soil properties using multivariate analyses. We expected to observe shifts in nutrient acquisition strategies along the dune sequence and a significant linkage of plant and fungal communities because of the plant–soil feedbacks described above.

MATERIALS AND METHODS

Study site

The study was conducted along a relic foredune plain known as 'Les Sillons' (Giles and King 2001). This sequence of dunes is located within the Îles de la Madeleine archipelago, in the southern Gulf of St Lawrence in Québec, Canada (47° 23' N, 61° 52' W). The region is characterized by a maritime cold temperate climate with a mean annual temperature of 4.5°C and annual average precipitation of 987 mm, of which approximately one-third falls as snow (Houle 2008). 'Les Sillons' is composed of a series of shore-parallel ridges forming a crescent-shaped area of 10.6 km². This dune system is a sandy depositional barrier that formed during the Holocene and now connects two bedrock islands (Giles and King 2001). This relic foredune plain is notable because it harbors a diversity of vegetation types over a short distance (generally <1 km), and includes a series of habitats ranging from the beach to forests (on dune crests) and wetlands (in dune swales) (Supplementary Fig. S1). The study site was divided into eight zones, moving away from the coastline: Z1, pioneer zone; Z2, front; Z3, back of the foredune; Z4, mobile dune; Z5, heathland; Z6, forest dune; Z7, inter-ridge swale; and Z8, lagoon wetland.

Field sampling

An extensive soil and vegetation field sampling was conducted in 'Les Sillons' in August 2011, in order to describe the plant and fungal communities associated with the different dune zones. We used a stratified sampling design by randomly selecting 10 sampling points from each of the eight zones for environmental data collection, vegetation surveys, and soil sampling within 1 m × 1 m plots. The location of the plots was determined prior to field sampling by using maps and aerial images to divide the sampling points throughout each zone. Each sampling plot is located at a minimal distance of 50 m from any other plot to reduce spatial autocorrelation. Slope, soil texture and organic layer thickness were determined at each sampling point. All vascular plants encountered in the plots were identified and their cover was estimated visually. The complete species list is

provided in Supplementary Table S1. In each plot, soil samples were composed of a mix of six soil cores of 0–25 cm depth that were collected randomly within the plot. The gravimetric water content was calculated as a ratio between the mass of water (weight of wet soil – weight of dry soil) and the dry weight of the sample. Conductivity, pH, extractable phosphorus (Mehlich-III), total phosphorus, carbon and nitrogen were measured using an air-dried portion of the soil samples. Soil nutrient concentrations were converted from a dry weight basis to a soil volume basis in order to make the values comparable across the different soil types, which included mineral sandy and organic soils. Soil properties are summarized in Supplementary Table S2, and details about the conversion to a soil volume basis can be found in Supplementary Appendix S1. Approximately 15 ml of previously homogenized soil samples were sub-sampled and frozen at -4°C within 6 h of sampling for subsequent molecular analysis.

Soil fungal community description

We characterized soil fungal communities by 454-sequencing of the internal transcribed spacer (ITS) region. The overall description of soil fungal diversity and community structure in this system was recently published (Roy-Bolduc et al. 2015), and the detailed molecular analysis and sequence processing methodology can be found in Appendix S1 of the supporting information. We used these fungal community data in some analyses presented here and will therefore quickly summarize the molecular and bioinformatics approach that was used. DNA was extracted from the soil samples and amplified using the ITS1F and ITS4 primers (White et al. 1990; Gardes and Bruns 1993). The amplicons were multiplexed, purified and sequenced using Roche 454 GS-FLX+ chemistry at the Genome Quebec Innovation Center (McGill University, Montreal, Canada). We initially retrieved 435 124 reads and retained 182 242 after quality filtering in QIIME (Caporaso et al. 2010) and Mothur (Schloss et al. 2009). Sequences were clustered into operational taxonomic units (OTUs) at a similarity level of 97% with Usearch v7.0 (Edgar 2010). Taxonomy was assigned to each OTU consensus sequence using the UNITE database (Kõljalg et al. 2013). Raw sequence data were deposited in the NCBI Sequence Read Archive and are available under the project number PRJNA269650.

Statistical analyses

All statistical analyses were conducted in R v3.0.2 (R Development Core Team 2014). The relative cover (%) of all vascular plants was computed by normalizing their absolute cover against total cover in each plot. We computed plant species richness as the number of species recorded in every given plot, and plant diversity using the Shannon and inverse Simpson diversity indices with the 'diversity' function in 'vegan' (Oksanen et al. 2016). Our cover data and sampling design (i.e. no sub-plots) did not allow us to use individual or sample-based rarefaction to provide a better estimate of species richness (Gotelli and Colwell 2001). We computed accumulation curves based on number of species per sample within each zone using the functions 'specpool' and 'poolaccum' in the 'vegan' library to ensure that our sampling design allowed us to representatively capture plant richness (Supplementary Fig. S2). We also assigned a nutritional strategy (facultative/obligate arbuscular mycorrhizal, ectomycorrhizal, ericoid mycorrhizal, arbutoid mycorrhizal, N-fixing, carnivorous, or unspecialized) to every plant species based on the literature (Supplementary Table S3). We

then computed the richness and diversity of the distinct plant nutritional strategies. Diversity was quantified using the Simpson index, which takes into account the abundance distribution of the different categories.

To examine patterns in fungal richness, we conducted OTU rarefaction analysis for individual samples. This analysis suggested that our sequencing depth was adequate since all curves reached or were close to reaching an asymptote and all samples had a Good's coverage value close to 1. Good's coverage and the Chao total richness estimator (Chao 1984) were both computed using the 'rarefaction.single' command in Mothur with 1000 iterations of re-sampling without replacement. Fungal diversity was evaluated the same way as plant diversity, using the Simpson and Shannon diversity indices. We tested the Pearson correlation between fungal and plant richness or diversity. We used one-way ANOVA and Tukey HSD post hoc tests to compare plants, nutritional strategies, and fungal richness and diversity among vegetation zones.

The relationship between plant and fungal community structure was tested using a co-inertia analysis. The analysis was performed on Hellinger-transformed data using the 'coinertia' function in the 'ade4' library (Dray and Dufour 2007) and significance of coefficients was tested with the 'randtest' function of the same package. The vector correlation coefficient (called the RV coefficient) is a multivariate generalization of the Pearson correlation coefficient that gives a measure of the consistency of the two data sets and takes a value between 0 and 1 (1 being perfectly identical).

Canonical redundancy analysis (RDA) based on Hellinger-transformed data was used to assess the influence of soil and topographic variables on community assemblages, using the 'vegan' library (Oksanen et al. 2016). The tests were performed on plant and fungal communities separately. Adjusted- R^2 values (Borcard, Gillet and Legendre 2011) were used to assess model fit. Statistical significance was assessed by permutation testing, using 999 permutations. The results of the RDAs were represented in a biplot using scaling type 2 in order to preserve the distances between sites. We also modeled community variance for plant and fungal communities with permutational analysis of variance (PERMANOVA) using the 'adonis' function in 'vegan'. The significance of the variance explained by the different parameters was tested with 999 permutations.

RESULTS

Plant and fungal community succession

Each of the eight zones sampled supported distinct vegetation (Fig. 1; Supplementary Table S2). The pioneer zone (Z1), fore-dunes (Z2 and Z3) and mobile dunes (Z4) consisted of younger sites located close to the shore, which were characterized by dry, poorly stabilized sandy soils and a plant community that was dominated by *Ammophila breviligulata* Fern., the American beachgrass. In areas that were not directly exposed to wind and salt spray, we also found a few herbaceous plants such as *Artemisia stellaria* L. and *Festuca rubra* L. Heathlands (Z5) are dry habitats and they were dominated by shrubs such as *Myrica pennsylvanica* Mirb., *Juniperus communis* L. and *Spiraea alba* Du Roi. This plant community transitioned progressively into black spruce, fir and pine forests (Z6). The inter-ridge depressions and wetlands (Z7 and Z8) harbored poorly drained organic soils and supported a diverse array of vegetation, including several Ericaceae, *Sphagnum* and *Carex* species. We detected significant differences in plant richness and diversity across the eight zones (Richness:

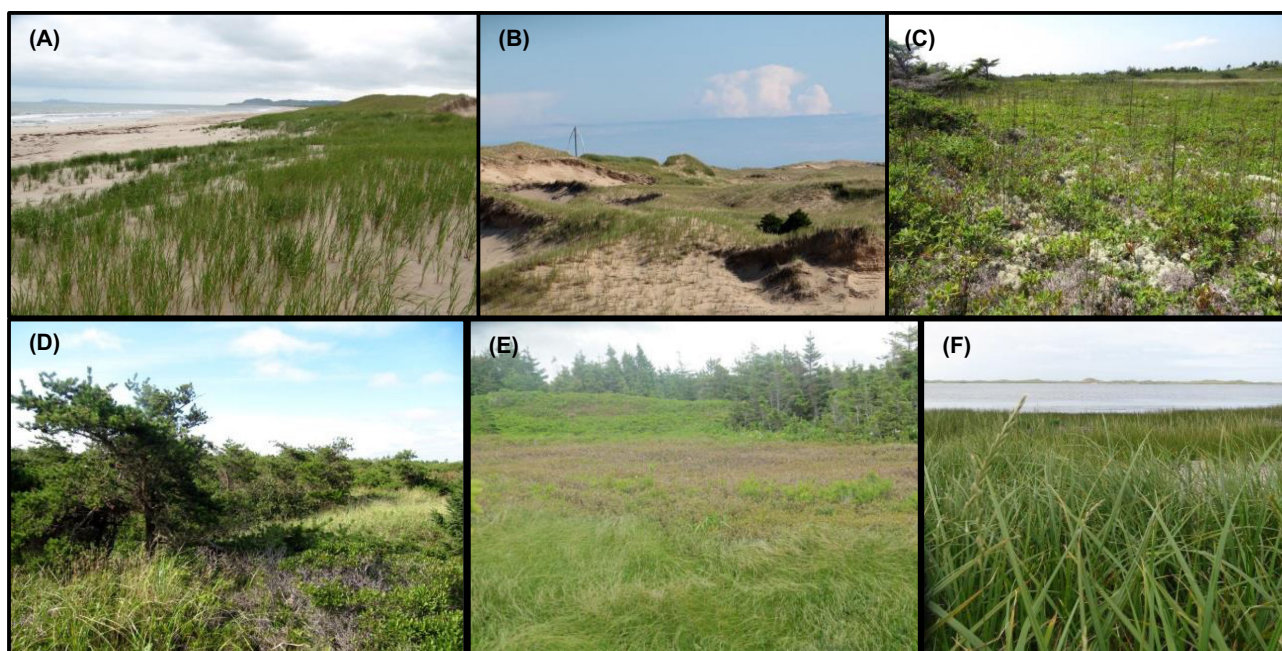


Figure 1. The different zones of our study site, from the pioneer zone and the foredune (A), to the mobile dune (B), the heathland (C) and the forested dune (D). At the end of the edaphic gradient, we can also observe wetlands in the inter-ridge swales (E) and the lagoon wetlands (F).

$F_{7,69} = 11.2$, $P < 0.001$; Shannon: $F_{7,69} = 15.14$, $P < 0.001$; Simpson: $F_{7,69} = 7.82$, $P < 0.001$). Moving from the coastline to the wetland, we observed a generally increasing diversity with the heathland and lagoon wetland being the most diverse zones (Table 1). Observed plant richness by sampling plot ranged between 1 and 14. A total of 73 plant species were identified across all the plots sampled.

With regard to soil fungi, functionally and taxonomically distinct communities were found in each zone of the successional and edaphic gradient, as previously described (Roy-Bolduc et al. 2015). Soil fungal richness was high across the succession gradient, with 3406 observed OTUs and a total richness of 4492 OTUs according the Chao estimator. Observed richness within individual sampling points ranged between 54 (in the pioneer zone) and 329 (in the lagoon wetland); however, there were no significant differences in either richness or diversity (Table 1). In addition, no significant correlations between fungal and plant richness nor alpha diversity were observed.

Plant nutrient-acquisition strategies

A clear increase in the number of strategies and their diversity was observed across the different zones (Fig. 2). The arbuscular mycorrhizal (AM) strategy was the only plant nutrient-acquisition strategy present in all zones and the plants using this strategy were dominant in terms of relative cover in the first four successional stages, i.e. from the pioneer zone (Z1) to the mobile dune (Z4; Fig. 3). Despite a high relative cover, AM plants displayed low richness levels in the first zones because of the dominance of *Ammophila breviligulata* Fern. (Supplementary Fig. S3). The ectomycorrhizal strategy was first observed in the mobile dune zone (Z4) and the relative cover of EM plants peaked in the forested zone (Z6). Even though it was represented by a low richness (Supplementary Fig. S3), its relative cover was high especially in the forested zone. Ericoid mycorrhizal plants were only found in the last zones, i.e. from the heathland (Z5) to the lagoon wetlands (Z8). Plants forming symbioses with nitrogen-

fixing bacteria were first observed in the mobile dune (zone 4) and their relative cover peaked in the inter-ridge swale (Z7). Plants without a specialized strategy were present mainly in the lagoon wetland, but also in zones 3, 5 and 7 but at very low densities (Fig. 3). Some strategies were rare and are therefore not shown in Fig. 3, such as carnivorous and arbutoid mycorrhizal (Supplementary Fig. S4). Two carnivorous plant species were identified in our sampling sites (*Drosera rotundifolia* L. and *Utricularia cornuta* Michx.) and were found mainly in the inter-ridge swale and in the lagoon wetland. *Arctostaphylos uva-ursi* L., a species from the Ericaceae forming arbutoid mycorrhizal symbiosis, was found in the heathland (Z5) and in the forested zone (Z6). Finally, we did not find any parasitic plants nor plants forming orchid mycorrhizas.

Linking plant and fungal communities

Despite the absence of a relationship between alpha diversity of plants and fungi, we observed coordinated changes in plant and soil fungal community structure. The RV coefficient measured in the co-inertia analysis was high (0.59) and highly significant ($P < 0.001$; after 9999 permutations), indicating a strong correlation between the two data sets (Fig. 4). Figure 4A shows the projection of the sampling sites in the co-inertia space with regards to both datasets. Sites from the first four successional stages are mainly located on the right of co-inertia plot, while heathland (Z5) and forested sites (Z6) are grouped in the top left and the humid sites (Z7 and Z8) at the bottom left. The first two axes represented, respectively, 73.85% and 7.12% of co-inertia. Arrows in panels B and C represent the importance and direction of the contribution of plant species and fungal genera to the distribution of sites in the co-inertia space. *Ammophila breviligulata* was the plant species with the most important contribution and was associated to the younger sites (Z1–Z4). *Juniperus communis*, *Myrica pensylvanica* and *Empetrum nigrum* were associated to Z5 and Z6, and *Myrica gale*, *Carex* sp., *Typha latifolia*, *Sphagnum* sp. and *Menyanthes trifoliata* to Z7 and Z8. Amongst fungi,

Table 1. Variation in plant and fungal alpha diversity across succession stages. Plant richness was measured as the number of observed species, while fungal richness was evaluated as the number of observed species, total richness Chao estimator, and rarefied richness (subsampling of 200 reads per sample). Diversity was measured with the Shannon and the inverse Simpson indices. Values are mean $\pm$ standard error ($n = 10$). Letters indicate significant differences between means assessed with Tukey HSD post hoc tests ($P \leq 0.05$). The last rows indicate the total richness, i.e. the total number of different plant species or fungal OTUs recorded in each zone.

	Aboveground vegetation							
	Z1	Z2	Z3	Z4	Z5	Z6	Z7	Z8
Pioneer zone								
Richness	1.00 $\pm$ 0.00 (a)	1.11 $\pm$ 0.11 (a)	2.44 $\pm$ 0.60 (ab)	2.6 $\pm$ 0.56 (ab)	6.44 $\pm$ 0.80 (cd)	4.56 $\pm$ 0.69 (abc)	5.20 $\pm$ 0.98 (bcd)	8.36 $\pm$ 1.30 (d)
Shannon	0.00 $\pm$ 0.00 (a)	0.07 $\pm$ 0.07 (a)	0.41 $\pm$ 0.18 (ab)	0.32 $\pm$ 0.12 (ab)	1.25 $\pm$ 0.11 (cd)	0.71 $\pm$ 0.16 (bc)	0.89 $\pm$ 0.18 (bcd)	1.39 $\pm$ 0.16 (d)
Simpson	1.00 $\pm$ 0.00 (a)	1.08 $\pm$ 0.08 (a)	1.66 $\pm$ 0.35 (ab)	1.29 $\pm$ 0.13 (a)	3.09 $\pm$ 0.43 (bc)	1.80 $\pm$ 0.29 (ab)	2.19 $\pm$ 0.32 (abc)	3.45 $\pm$ 0.56 (c)
Total richness	1	2	10	12	28	23	23	43
Soil fungi								
Observed richness	107.30 $\pm$ 14.59	121.44 $\pm$ 9.74	151.00 $\pm$ 15.57	150.00 $\pm$ 22.80	181.67 $\pm$ 24.93	173.56 $\pm$ 22.69	152.80 $\pm$ 21.50	164.73 $\pm$ 23.07
Rarefied richness	99.28 $\pm$ 8.60	106.40 $\pm$ 6.78	112.40 $\pm$ 11.13	114.32 $\pm$ 13.17	124.93 $\pm$ 11.02	125.28 $\pm$ 13.62	118.32 $\pm$ 16.05	118.44 $\pm$ 12.88
Chao total richness	129.83 $\pm$ 18.24	148.06 $\pm$ 12.23	182.88 $\pm$ 18.65	187.38 $\pm$ 28.79	227.61 $\pm$ 30.65	223.29 $\pm$ 26.05	183.36 $\pm$ 26.18	198.94 $\pm$ 26.64
Shannon	3.16 $\pm$ 0.20	3.31 $\pm$ 0.18	3.28 $\pm$ 0.25	3.33 $\pm$ 0.23	3.35 $\pm$ 0.25	3.15 $\pm$ 0.31	3.14 $\pm$ 0.29	2.92 $\pm$ 0.32
Simpson	12.78 $\pm$ 2.36	13.55 $\pm$ 2.58	13.12 $\pm$ 2.72	14.05 $\pm$ 2.92	15.55 $\pm$ 4.49	13.23 $\pm$ 4.44	13.18 $\pm$ 3.87	11.38 $\pm$ 3.50
Total richness	607	519	685	892	1039	920	942	1157

Sordariomycetes spp. made the most important contribution to site distribution and was associated with humid sites (Z7 and Z8).

Influence of soil physicochemical properties on community structure

Humus thickness, gravimetric water content and organic carbon significantly increased along the successional and edaphic gradient, while pH and extractable phosphorus generally decreased (Supplementary Table S1). We observed a significant correlation between soil properties and plant community structure (Fig. 5A). Our model explained 39.2% (adjusted R^2) of the variation in plant community ($F_{7,69} = 7.99$; $P < 0.001$). Soil fungal community was also linked to soil properties but to a lesser extent, the model explaining 11.4% (adjusted R^2) of variation in soil fungal community composition ($F_{7,69} = 2.40$; $P < 0.001$) (Fig. 5B). The two first canonical axes explained 39.1% and 12.8% of the total model variance for plant and fungal communities, respectively. Although soil physicochemical properties had a more important effect on vegetation (as determined by a higher adjusted R^2 value), their influence on plant and fungal communities was qualitatively similar (Table 2). Soil pH had the highest R^2 value in both models. The proportion of variation explained was significant for humus thickness, pH, soil water content, total nitrogen and total carbon for fungi. Neither total nor extractable phosphorus explained a significant proportion of the variation in plant and fungal community composition.

DISCUSSION

Our comprehensive sampling of the vegetation, combined with the characterization of soil fungi using high-throughput sequencing, allowed us to explore changes in soil microbial and plant communities along a relic foredune plain forming a successional and edaphic gradient. We observed strong linkage between plant and fungal communities. Our results also showed a clear increase in plant richness and an increasing diversity of nutrient-acquisition strategies along the sequence. Finally, we found a significant correlation between plant and fungal communities, which were influenced by similar soil physicochemical properties.

First, we detected distinct plant and soil microbial communities across the successional and edaphic gradient. Given the contrasted abiotic conditions existing in the different dune zones (e.g. in terms of wind, salinity and soil water content), variation in plant community composition was expected (Gurevitch, Scheiner and Fox 2002). We also expected that changes in soil fungal community structure would be strongly linked to vegetation dynamics (Roy-Bolduc et al. 2015). Indeed, a co-inertia analysis produced a high and significant RV coefficient, demonstrating a strong correlation between plant and fungal community structure across the successional and edaphic gradient. These results are consistent with the current knowledge on the feedbacks existing between plants, microbes and soil (e.g. Wardle et al. 2004; Barberán et al. 2015). Vegetation, on the one hand, is largely acknowledged as a key driver of soil microbial communities (e.g. Prescott and Grayston 2013). Plants are hosts to many associations, both symbiotic and parasitic, with fungi (Broeckling et al. 2008; Barberán et al. 2015). Also, because saprophytic fungi are the main decomposers of plant dead material and rely directly on plant products such as leaf litter and root exudates, their community composition is often influenced by vegetation (Wardle et al. 2004; Boddy, Frankland and van West 2007;

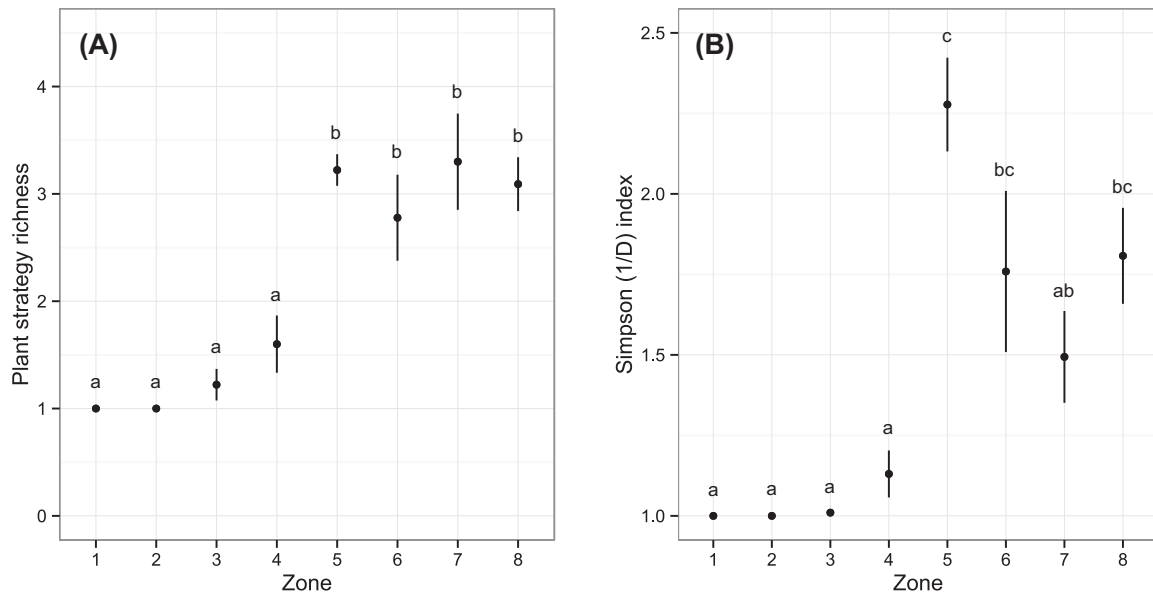


Figure 2. Richness (A) and diversity (B) of plants nutrient-acquisition strategy by sampling plots along the successional and edaphic gradient. Obligate and facultative AM are considered as the same strategy, and unknown were removed from the dataset for this analysis. Points are means and bars standard errors ($n = 10$). Different letters indicate significant differences ($P \leq 0.05$).

Miniati et al. 2007). On the other hand, soil microbes have an impact on plant productivity, diversity and community composition through various processes implying direct (root-associated) or indirect (free-living) interactions that can be either positive or negative (van der Heijden, Bardgett and van Straalen 2008). For example, soil pathogens such as *Fusarium* spp. can reduce plant productivity (a direct negative effect), while free-living soil saprotrophs are able to enhance soil nutrient availability by breaking down organic matter into inorganic forms available to plants (indirect positive effect) (Boddy, Frankland and van West 2007; van der Heijden, Bardgett and van Straalen 2008). Several fungi involved in such interactions with plants were recorded in this dune system (Roy-Bolduc et al. 2015). For example, the arbuscular mycorrhizal genus *Glomus* was often recorded in the foredune where it most likely associates to *Ammophila breviligulata* and improves its nutrition, growth and survival (Smith and Read 2008). The occurrence of other symbiotic fungi such as the EM *Sclerotinia*, *Thelephora*, *Cortinarius* and *Clavulina* indicates potential positive interactions occurring between fungi and their host plants. We also observed pathogenic fungi such as *Enthorrhiza* that might limit the growth of *Juncus* and *Cyperaceae* in the wetlands. Finally, several saprotrophs (e.g. *Dendryphiella*, *Emicellopsis*, *Neobulgaria* and *Schizothecium*) were observed along the dune sequence and might have indirect beneficial impact on plants by improving nutrient availability. The results of the co-inertia analysis did indicate strong linkage of plant and fungal communities by connecting presence of members from the two groups. However, further experimental studies would be required to go beyond these co-occurrence patterns and to verify realized interactions and describe their ecological impacts and functions.

Second, we observed a significant increase in plant richness and diversity along the gradient. This pattern of increasing plant diversity with soil age has been observed in several chronosequences under various climates (Laliberté et al. 2013). For example, Wardle et al. (2008) studied tree and plant diversity through succession and ecosystem retrogression (i.e. the decline phase of succession that is characterized by decreasing soil fer-

tility and plant biomass) at six different locations. They found a significant increase in vascular plant diversity at all locations but one, and observed an increasing number of plants with smaller statured life forms. Zemunik et al. (2015) also observed a clear increase in plant richness and diversity along a well-studied Australian dune chronosequence. Possible explanations for this pattern of increasing plant diversity with greater soil age are the decline of competition intensity among co-occurring plants during ecosystem retrogression, greater spatial heterogeneity of soil nutrients, and increased light availability because of fewer trees (Wardle et al. 2008). However, most of these mechanisms remain incompletely understood and still need to be tested empirically (Laliberté et al. 2013). Our data do not allow us to describe the mechanisms responsible for the pattern of plant diversity we observed. Still, the complementarity in belowground functional traits linked to resource acquisition along the successional gradient, which coincided with increasing plant richness and diversity, is consistent with results of a recent study (Zemunik et al. 2015) and suggests a potential importance of belowground resource partitioning (Laliberté et al. 2013). Indeed, access to distinct nutrient pools and diverse nutrient-uptake strategies can reduce interspecific competition for soil resources and thus promote local plant coexistence (Turner 2008; Laliberté et al. 2013). Complementarity of nutritional strategies could also perhaps facilitate the growth of neighbor plant species, which might promote plant species diversity. For example, Teste et al. (2014) observed the best growth of *Melaleuca* seedlings in the presence of both ectomycorrhizal host plants and nutrient mining plants with cluster roots, suggesting a synergistic effect between the two strategies. The trends we observed in nutrient-acquisition strategies are consistent with the model proposed by Lambers et al. (2008), where AM plants are more abundant in young soils, followed by ectomycorrhizal, ericoid mycorrhizal and then non-mycorrhizal plants. The observation of AM as the dominant nutrient uptake strategy is consistent with the fact that the majority of plant species associate with AM fungi (Smith and Read 2008). Zemunik et al. (2015) also identified the AM strategy as the dominant one along a dune chronosequence. Most plant

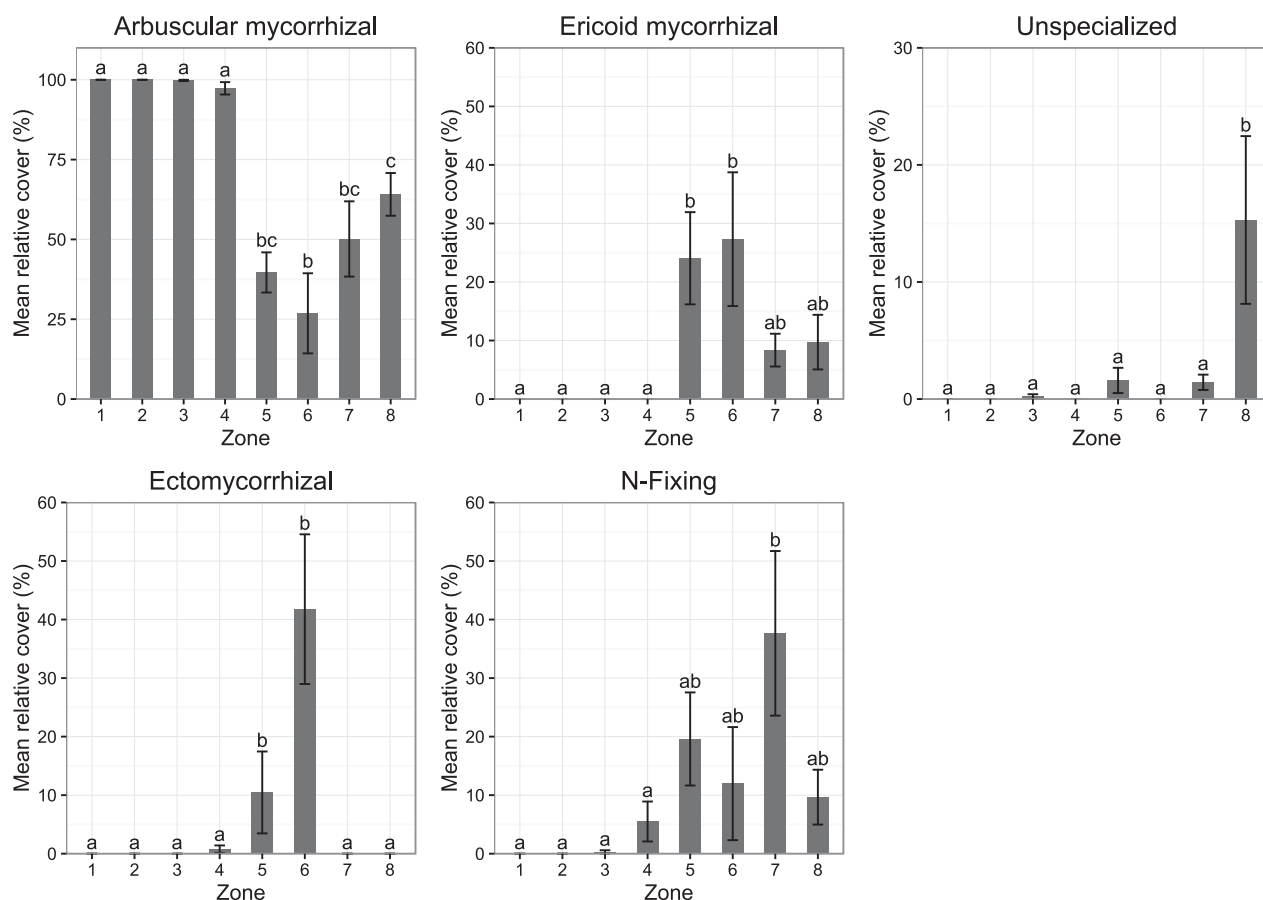


Figure 3. Mean relative cover of the main nutrient-acquisition strategies along the successional and edaphic gradient. Bars represent standard errors ($n = 10$) and different letters indicate significant differences ($P \leq 0.05$).

nutrient acquisition strategies linked to fungal symbionts followed the distribution of associated fungal functional guilds. Ectomycorrhizal fungi and ericoid mycorrhizal fungi displayed high abundance in Z5 and Z6 (Roy-Bolduc et al. 2015), a pattern similar to the associated plant nutritional strategy. However, occurrences of EM and ericoid fungi were also recorded in the coastal zone without the presence of associated hosts (Roy-Bolduc et al. 2015). That might be explained by the fact that many of these fungi are facultative saprotrophs (Tedersoo et al. 2003; Cullings and Courty 2009). The distribution and abundance of AM fungi seemed limited to Z2 and Z3, which contrast with the fact that AM plants were present in all zones of the dune sequence. However, ITS markers do not allow us to adequately capture Glomeromycota diversity and the observed abundance of AM fungi is most likely an underestimation (Lee, Lee and Young 2008).

We also found significant correlations of both plant and fungal community composition to soil physicochemical properties. Of all the soil properties investigated, pH was the one explaining the highest proportion of variance in plant and fungal community composition. Soil pH has previously been identified as one of the key drivers of plant community composition along an Australian dune chronosequence (Zemunik et al. 2015; Laliberté, Zemunik and Turner 2014). Even though fungi are thought to have wide pH optima (Rousk et al. 2010), some fungal groups such as arbuscular mycorrhizal fungi are described as highly responsive to variations in soil pH (Dickie et al. 2013),

which could have an indirect effect on the vegetation. Also, the impact of soil pH on nutrient solubilization and availability can explain the extent of its influence on plant communities (Gobat, Aragno and Matthey 2004). Soil water content is another property that we found to be influential, as expected. Cornwell and Ackerly (2009) also found that gravimetric soil water content strongly affected plant functional trait distribution. For example, they predicted an increase in plant richness following increasing soil water, as we observed here. Also, fungi are known to be very sensitive to anoxic conditions (Lin et al. 2012; Bragazza et al. 2015). Water content was the second most important environmental factor explaining variation in fungal community structure. The various and complex factors shaping plant diversity along a gradient, such as the one we studied here, remain unresolved. One potential mechanism, namely environmental filtering, explains local plant richness by the filtering of the species poorly adapted to local conditions from the regional species pool. Previous studies suggested that environmental filtering driven by soil acidification was responsible for plant assembly across a dune chronosequence (Laliberté, Zemunik and Turner 2014; Zemunik et al. 2015). Similarly, our results also indicate that soil pH could act as an environmental filter along this relic foredune plain. Furthermore, we detected similar responses of plants and fungi to the same soil parameters. This could be the result of indirect interactions between above- and below-ground communities through modification of the soil environment. In other words, vegetation might be influencing (at least

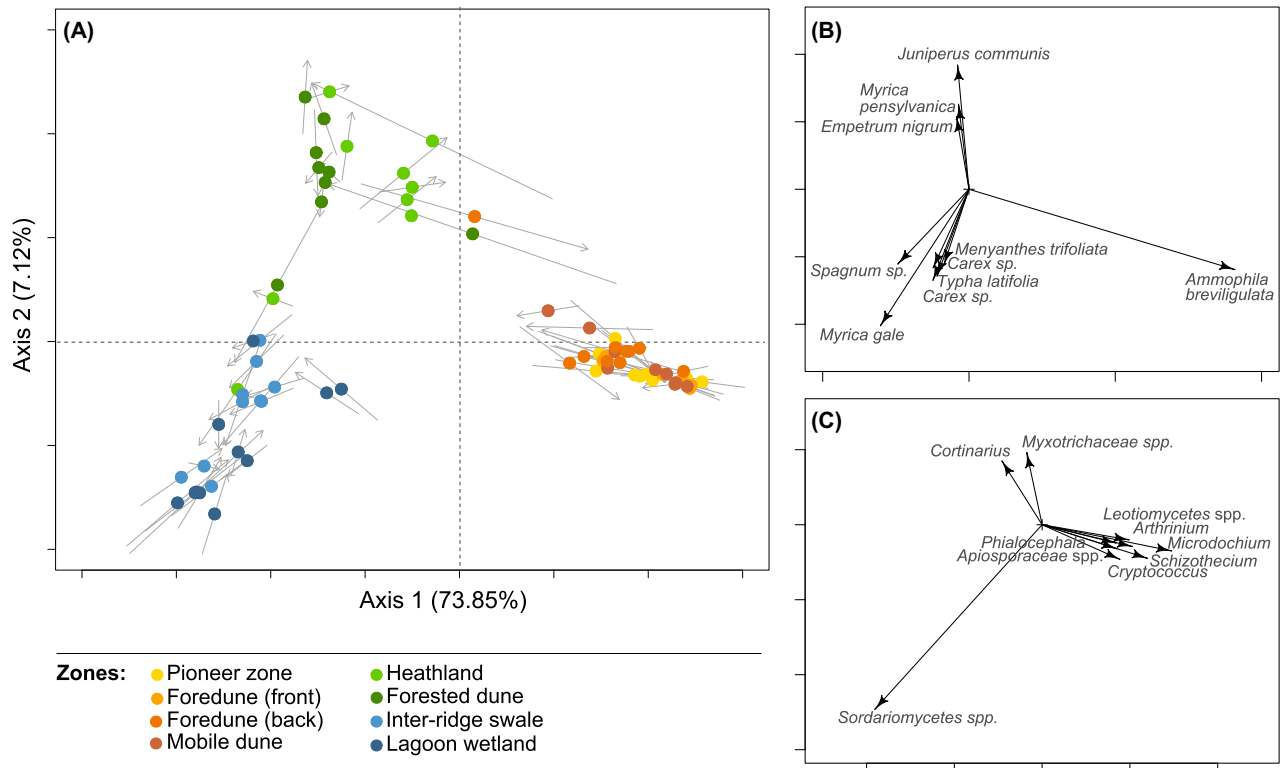


Figure 4. Joint site plot (A) depicting consensual ordination of sites as a function of both plant and fungal community compositions preserving the Hellinger distances, and projection of the 10 plant species (B) and fungal genera (C) with the highest scores. The two first canonical axes account for 80.97% of total co-inertia and the RV coefficient is 0.59 ($P = 0.001$). Arrow length is proportional to the difference between the ordinations of the two data sets: position of the arrow tails is determined by the ordination of plant community data, and the head by fungal community data.

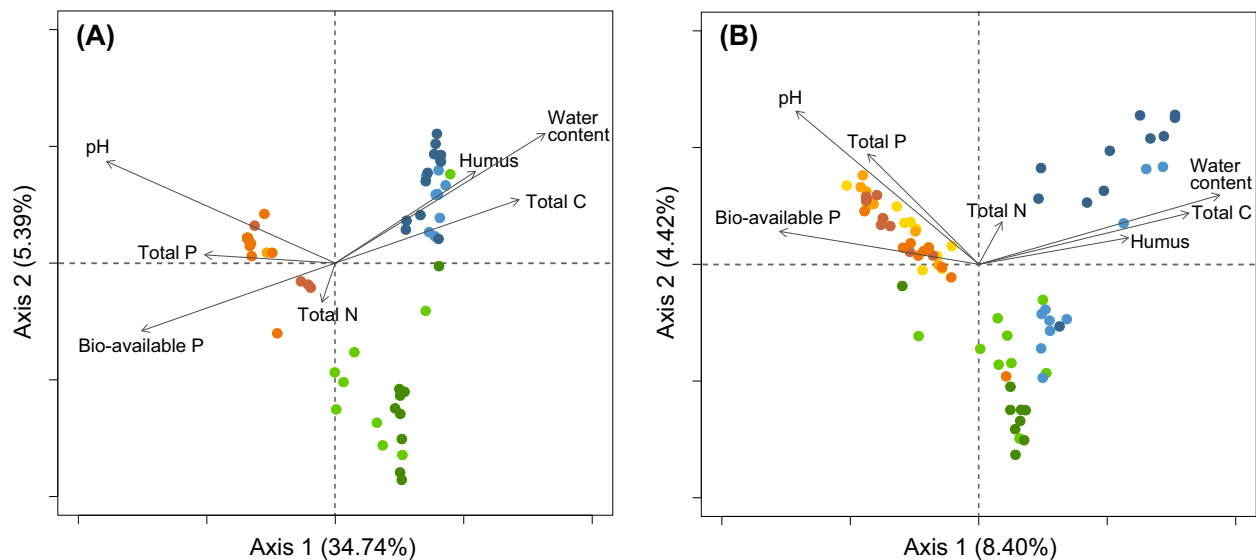


Figure 5. Transformation-based RDA biplots of plant (A) and fungal (B) communities in relation to soil properties using the Hellinger distance. Scaling type = 2. Adjusted R^2 is 0.3916 ($P = 0.001$) for the analysis on vegetation and 0.1142 ($P = 0.001$) for fungi.

partially) soil microbiota through modification of soil properties. However, our data do not allow us to discriminate between this hypothesis and that of similar independent reactions to environmental variables.

In conclusion, our study highlighted the strong linkage between plants and the soil biota in a nutrient-poor successional environment. The patterns we observed in nutrient-acquisition

strategies indicate that symbiotic interactions with fungi have an influence on vegetation dynamics. As the positive effects of microbes are known to be most common in nutrient-poor habitats where they enhance the supply of growth limiting nutrients (Jonsson et al. 2001; van der Heijden, Bardgett and van Straalen 2008; Teste et al. 2014), plant-fungi symbiotic interactions could be important for plant establishment and survival

Table 2. Community variance (portion of the model non-adjusted R^2) explained by soil physicochemical properties. Significant relationships (at $P \leq 0.05$) are in bold.

	Plant community		Fungal community	
	R^2	P-value	R^2	P-value
Humus thickness	0.11458	0.001	0.03766	0.001
pH	0.20532	0.001	0.05063	0.001
Water content	0.07731	0.001	0.04830	0.001
Total carbon	0.01459	0.085	0.01722	0.022
Total nitrogen	0.01698	0.040	0.01673	0.018
Total phosphorus	0.01232	0.141	0.01289	0.235
Extractable phosphorus	0.00658	0.548	0.01238	0.291
Residuals	0.55232		0.80418	
Total	1.00000		1.00000	

in this relic foredune plain. The diversification of belowground traits promoted functional diversity, and might favor resource partitioning and facilitation among co-occurring plant species, which could be one of the mechanisms favoring local plant diversity (Zemunik et al. 2015). Hence, local plant community assembly could be partly determined by mechanisms of resource competition and/or facilitation between co-occurring plant species. On the other hand, we also observed that soil properties, such as pH and water content, shape community structure, which suggests that environmental filtering is also important. Our results suggest that both environmental filtering (exclusion of species outside their viable range) and limiting similarity (coexistence of functionally dissimilar species due to competition for limiting resources) can together drive plant community assembly. Our finding that changes in soil microbial and plant communities were strongly coordinated along this sequence highlights the importance of aboveground–belowground linkages during ecological succession (Walker and Wardle 2014). Such aboveground–belowground linkages should be considered in conservation and ecological restoration practices (Kardol and Wardle 2010). This can be particularly important for ecosystems that are vulnerable to disturbance, such as coastal sand dunes.

SUPPLEMENTARY DATA

Supplementary data are available at FEMSEC online.

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Conflict of Interest. None declared.

Data Accessibility

The raw .sff files from 454 sequencing have been deposited in the Sequence Read Archive with Accession no. SRP059280. Available at: <http://www.ncbi.nlm.nih.gov/Traces/sra/?study=SRP059280>.

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